

ECOCIDE

Kakhovka Dam destruction: satellite vegetation and flood evidence

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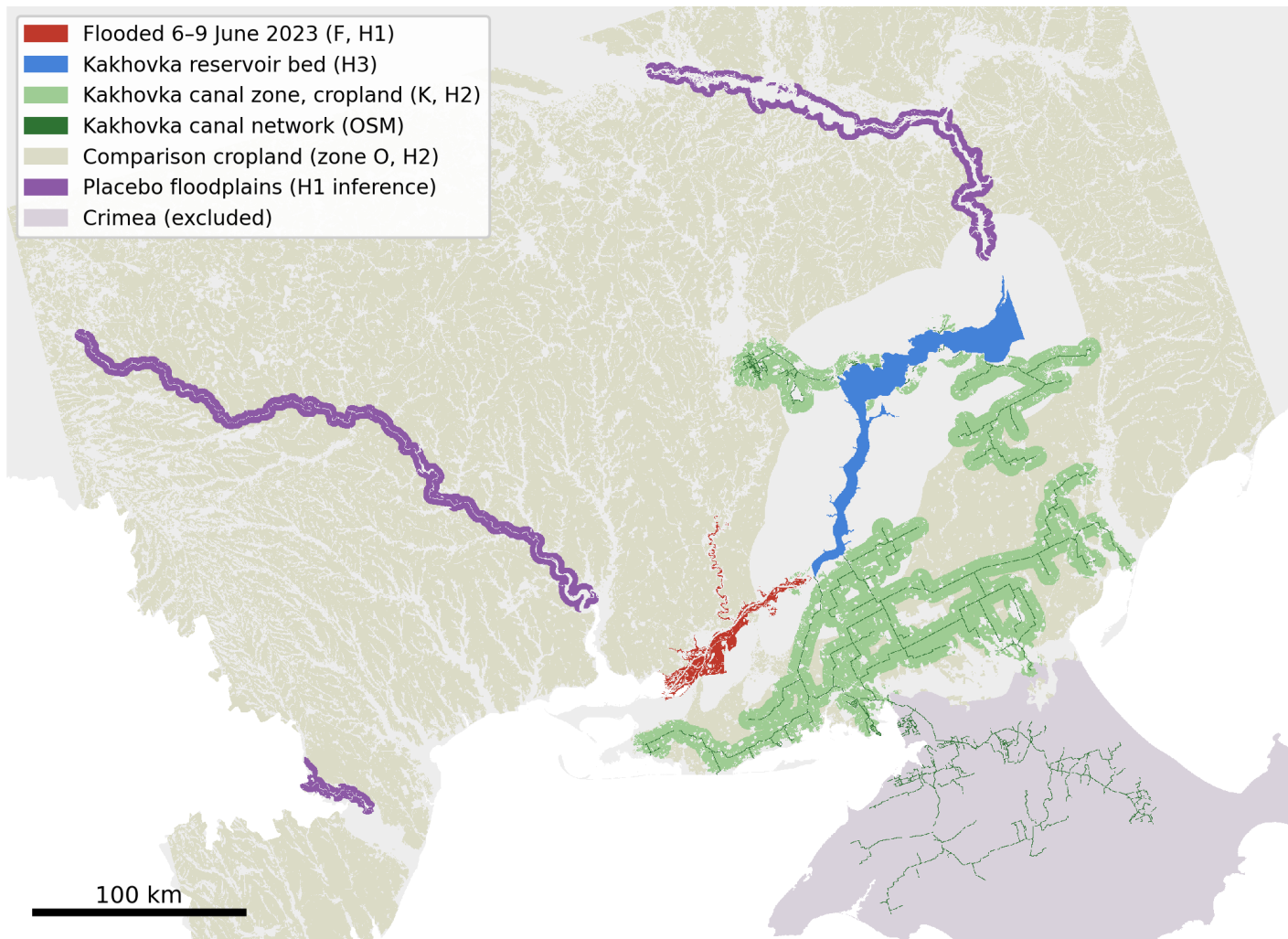
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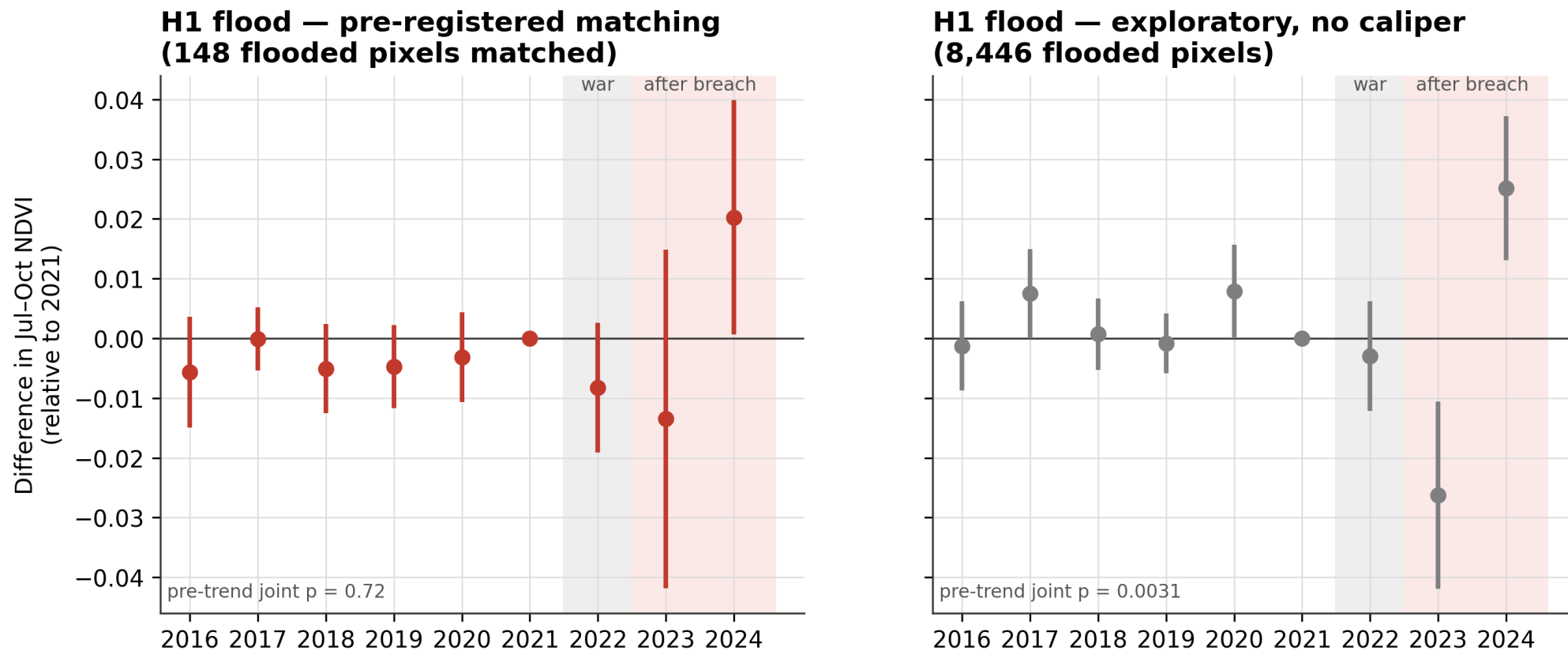
Figure S2-1 — Study 2 exposure groups

Study 2 exposure groups on the 231 m grid (EPSG:3035)



Flooded land (UNOSAT 6-9 June 2023 composite), former reservoir bed, Kakhovka canal zone (K) and comparison cropland (O), placebo floodplains; 231 m grid, EPSG:3035.

Figure S2-2 — H1 flood: event studies



Flooded minus matched unflooded pixels, relative to 2021. Left: pre-registered caliper matching; right: exploratory matching without caliper. 95% cluster-robust CIs.

Figure S2-3 — H2 irrigation: event study (pre-registered)

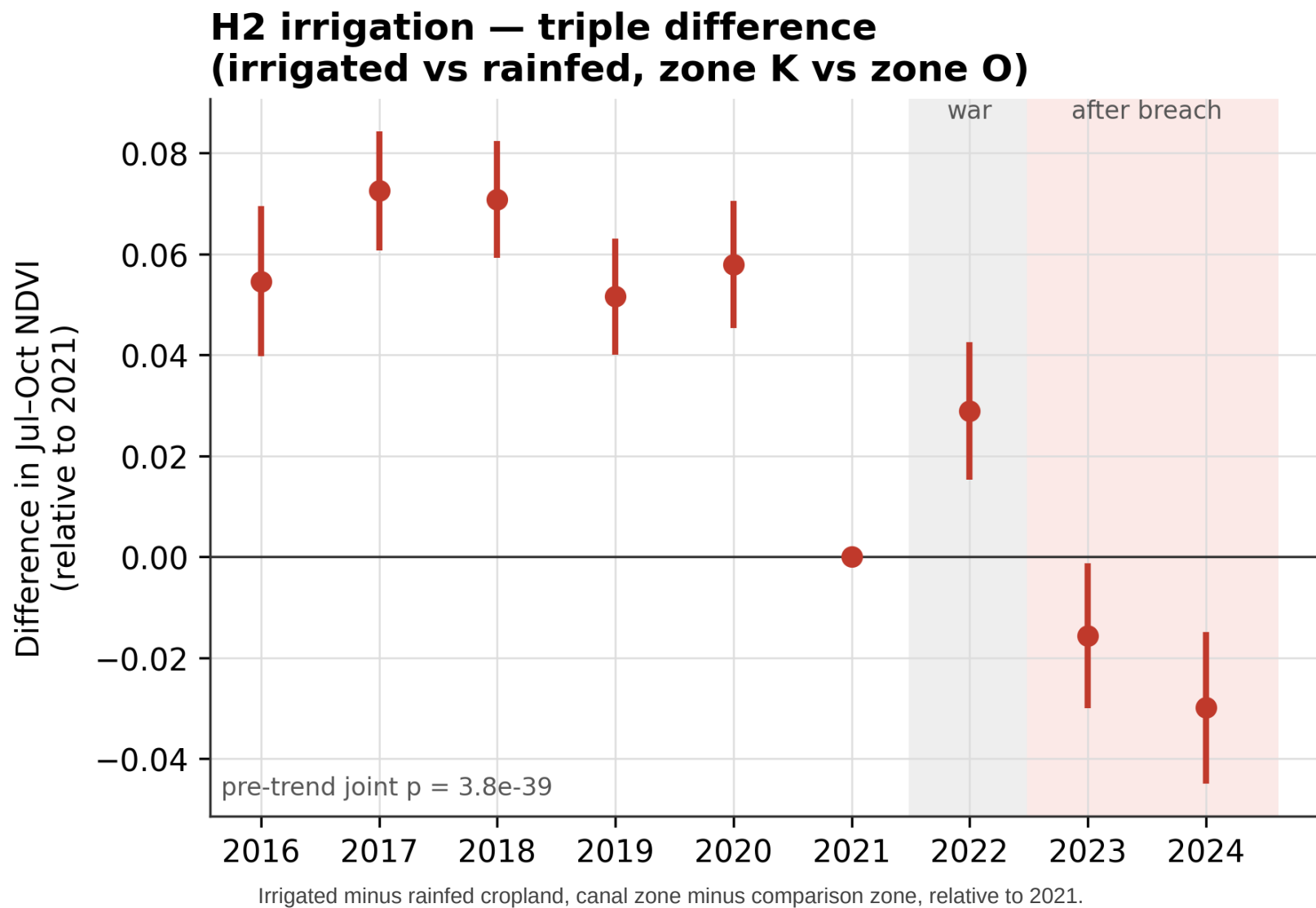
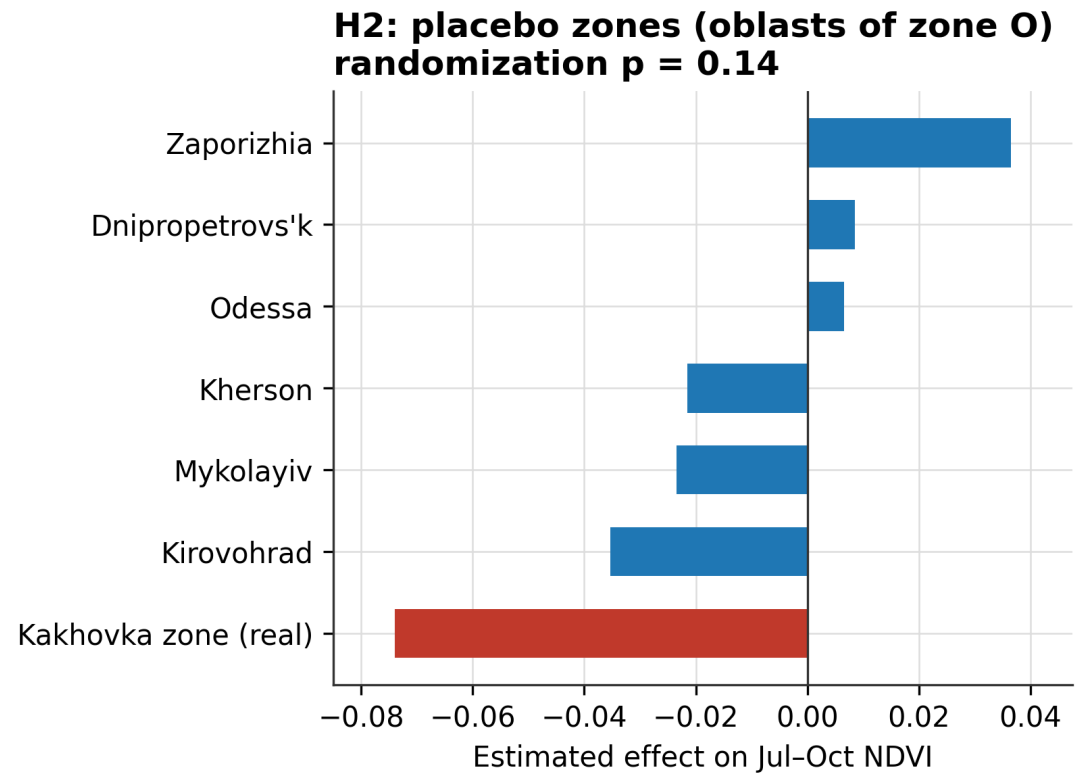
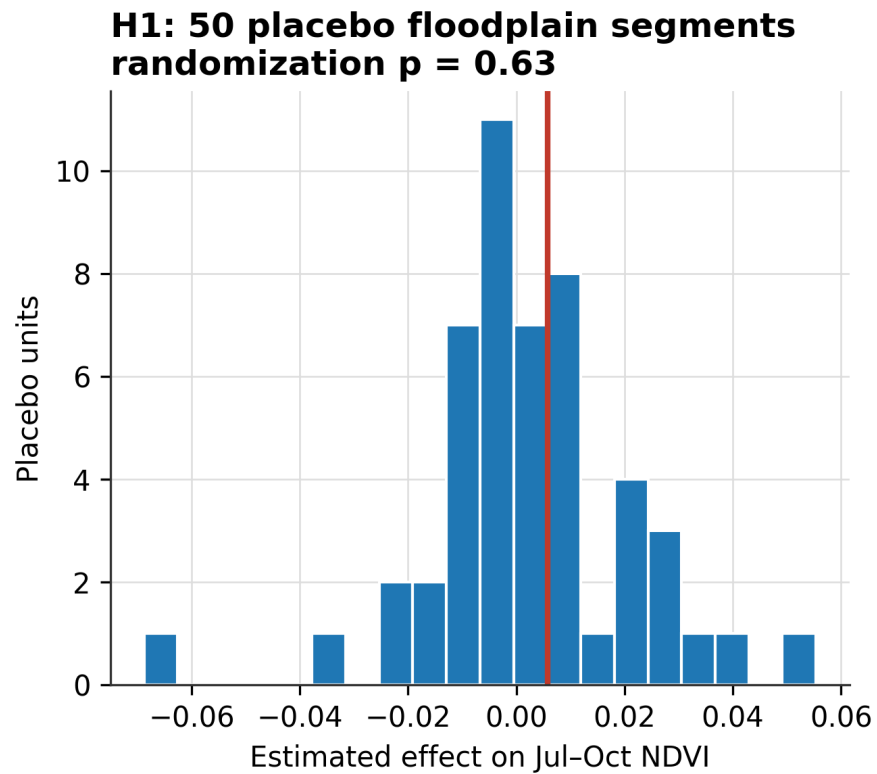
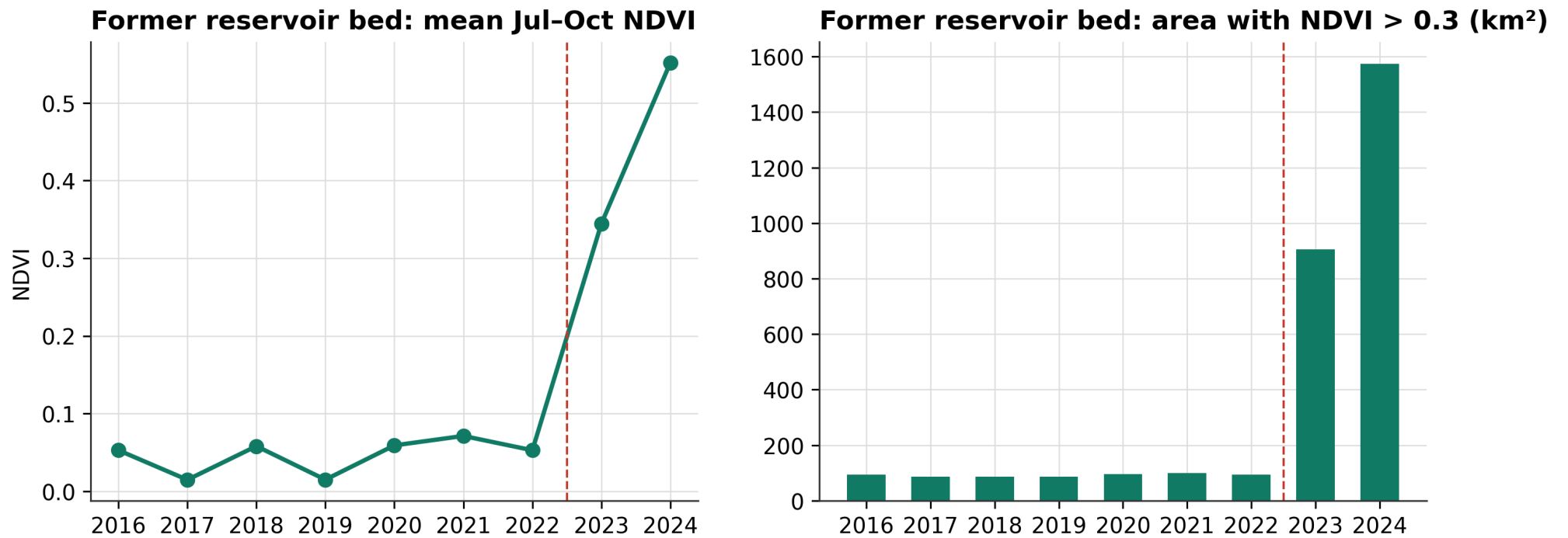


Figure S2-4 — Randomization inference



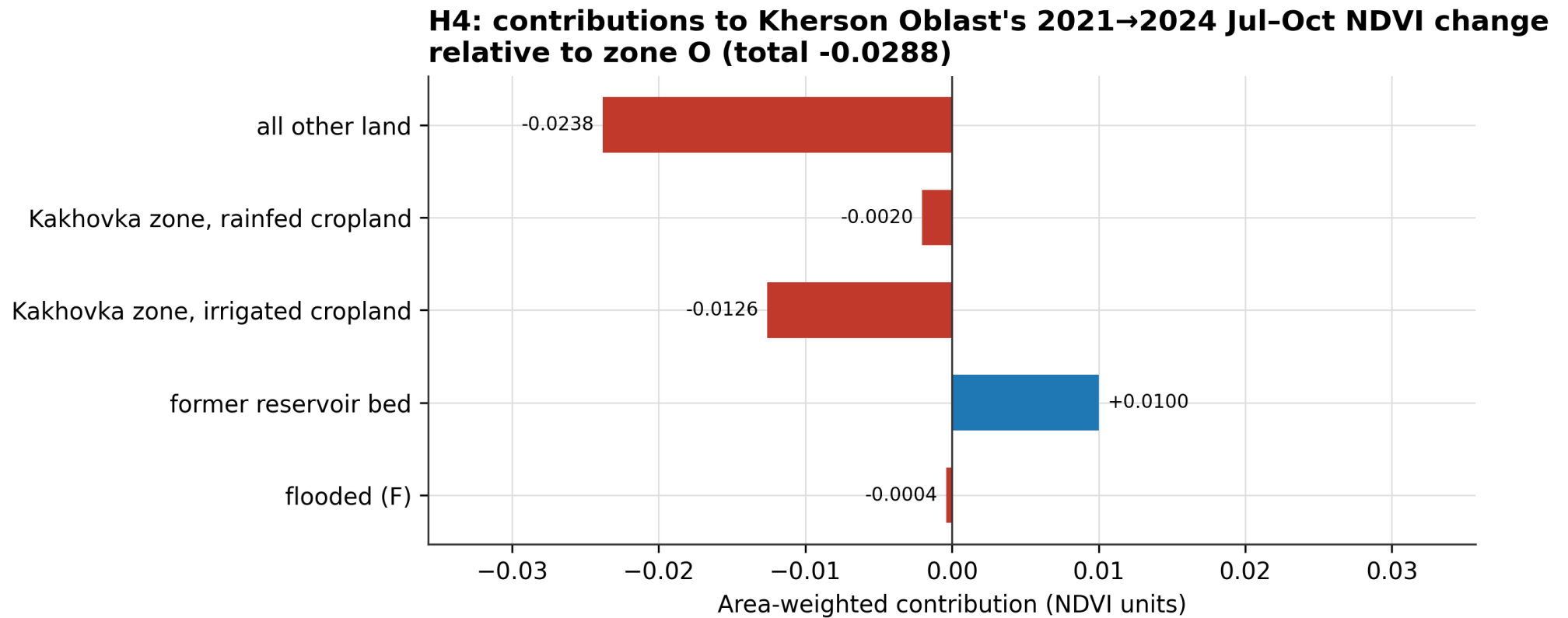
H1 estimate against 50 placebo floodplain segments; H2 estimate against six placebo oblasts.

Figure S2-5 — H3 former reservoir bed



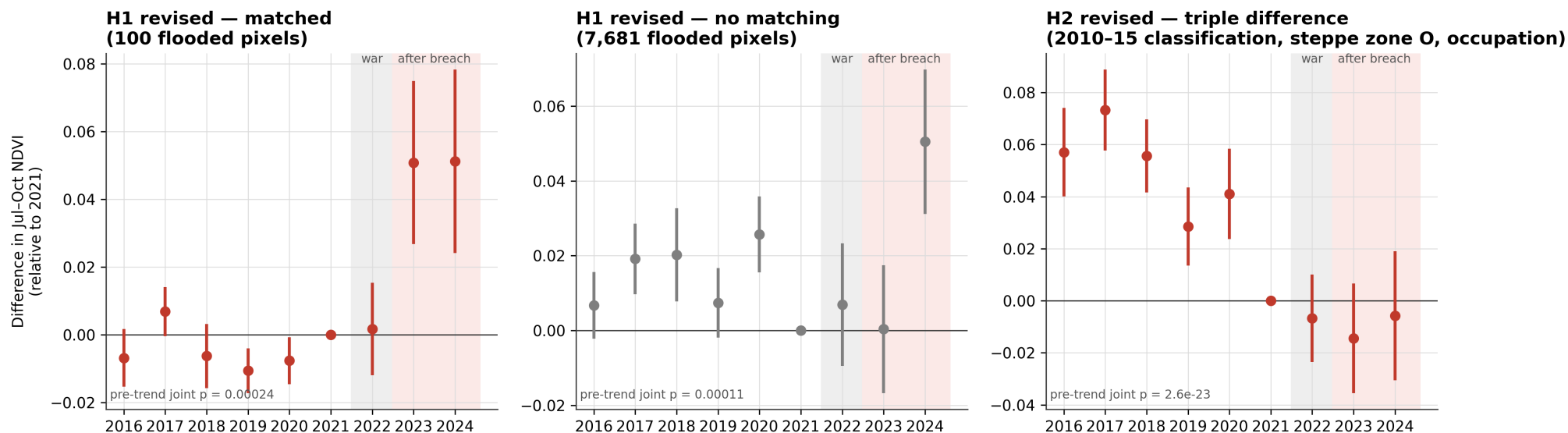
Mean July-October NDVI and area with NDVI above 0.3 on the former Kakhovka reservoir bed, 2016-2024.

Figure S2-6 — H4 decomposition of Kherson's change



Area-weighted contributions to Kherson Oblast's 2021-2024 change in July-October NDVI relative to zone O.

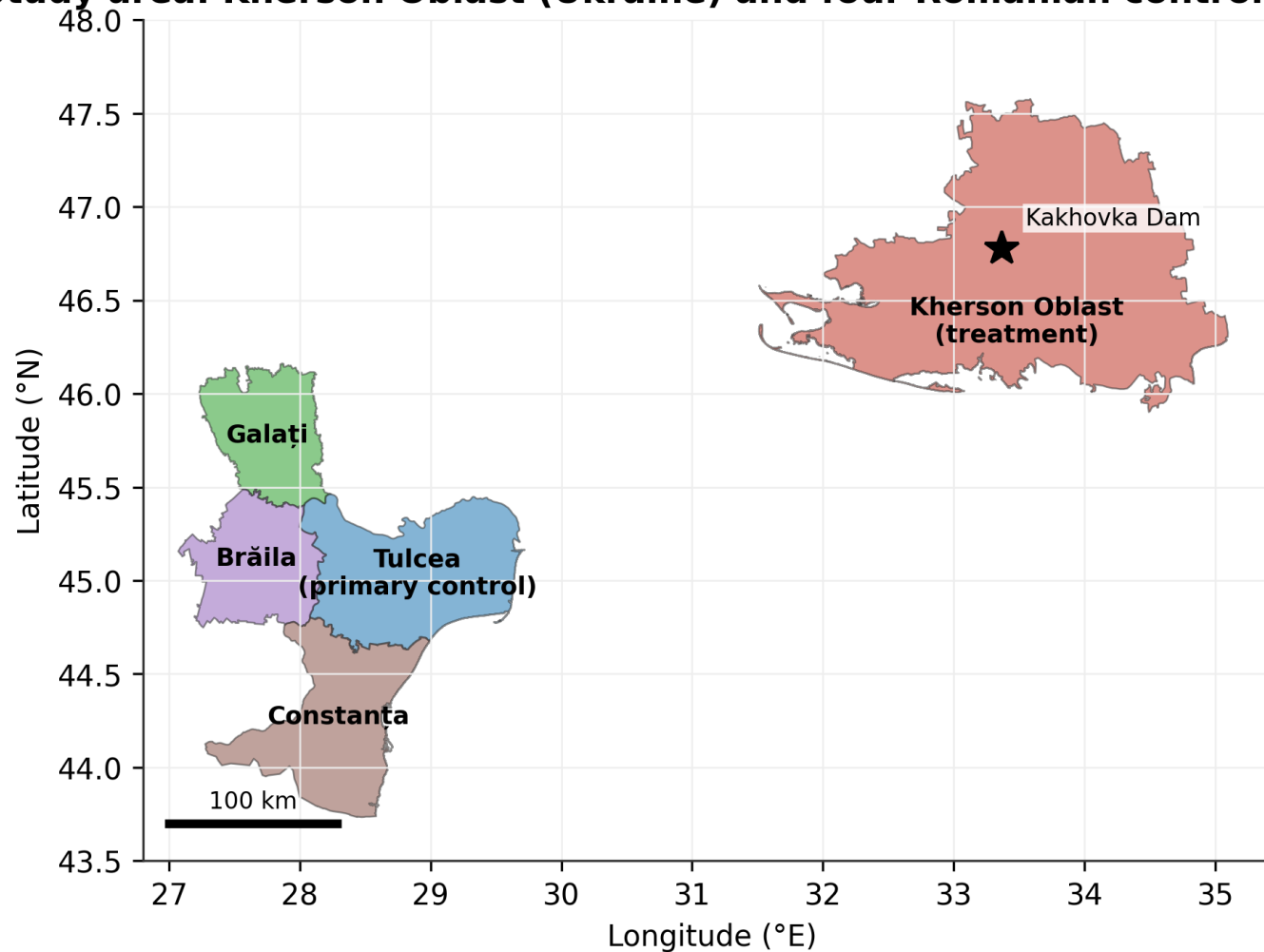
Figure S2-7 — Registered revision: event studies



H1 matched and unmatched, and H2 triple difference under Registered Revision 1 (2010-15 irrigation classification, steppe comparison zone, VIINA occupation and conflict, annual water masks).

Figure S1-1 — Study area

Study area: Kherson Oblast (Ukraine) and four Romanian control counties



Boundaries: GADM v4.1. Tulcea borders Ukraine's Odesa Oblast along the Danube.

Treatment zone (Kherson Oblast, Ukraine) and the four Romanian control counties (Tulcea, Galati, Braila, Constanta) in their true geographic positions. Boundaries: GADM v4.1.

Figure S1-2 — Lower Dnipro before the breach (April-May 2023)

BEFORE (MAY 2023)



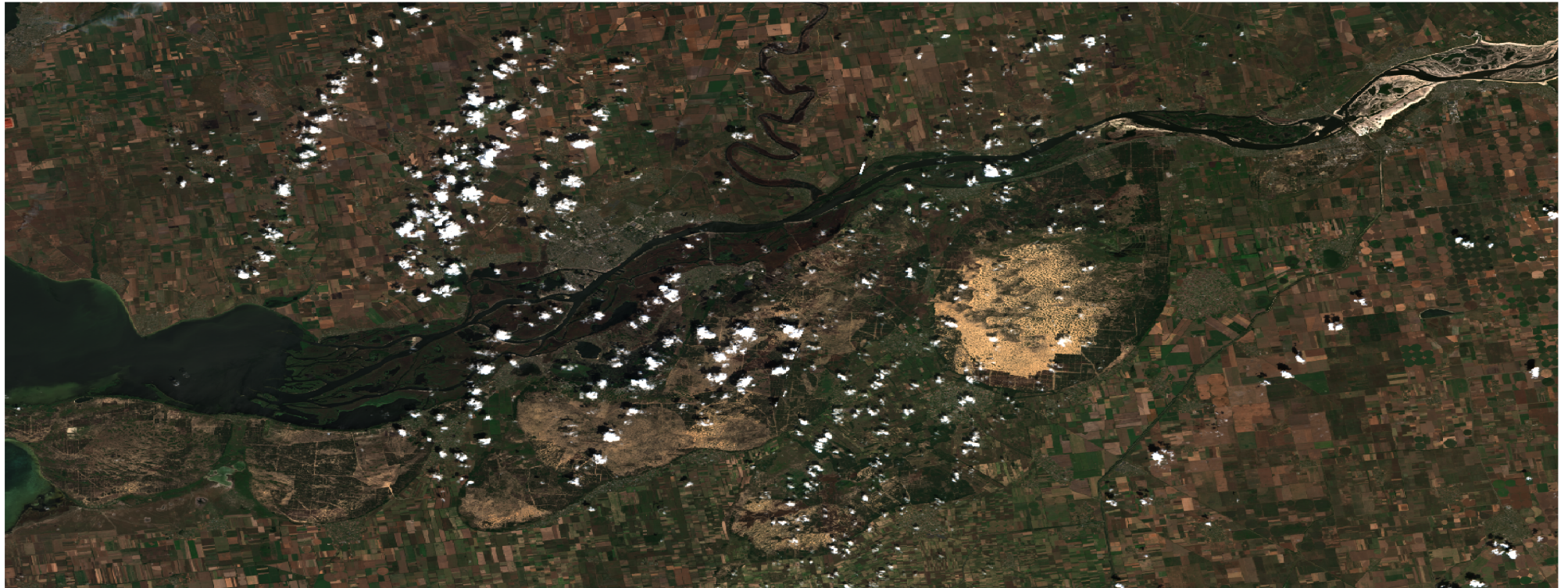
0 30 60 km

A horizontal scale bar with a black segment from 0 to 30 km and a white segment from 30 to 60 km.

Sentinel-2 L2A true-colour mosaic, 1 April - 31 May 2023, least-cloud mosaicking, bbox 32.0-33.6E, 46.3-46.9N. The frame covers the downstream floodplain and only the south-western tip of the reservoir.

Figure S1-3 — Lower Dnipro after the breach (July 2023)

AFTER (JULY 2023)



0

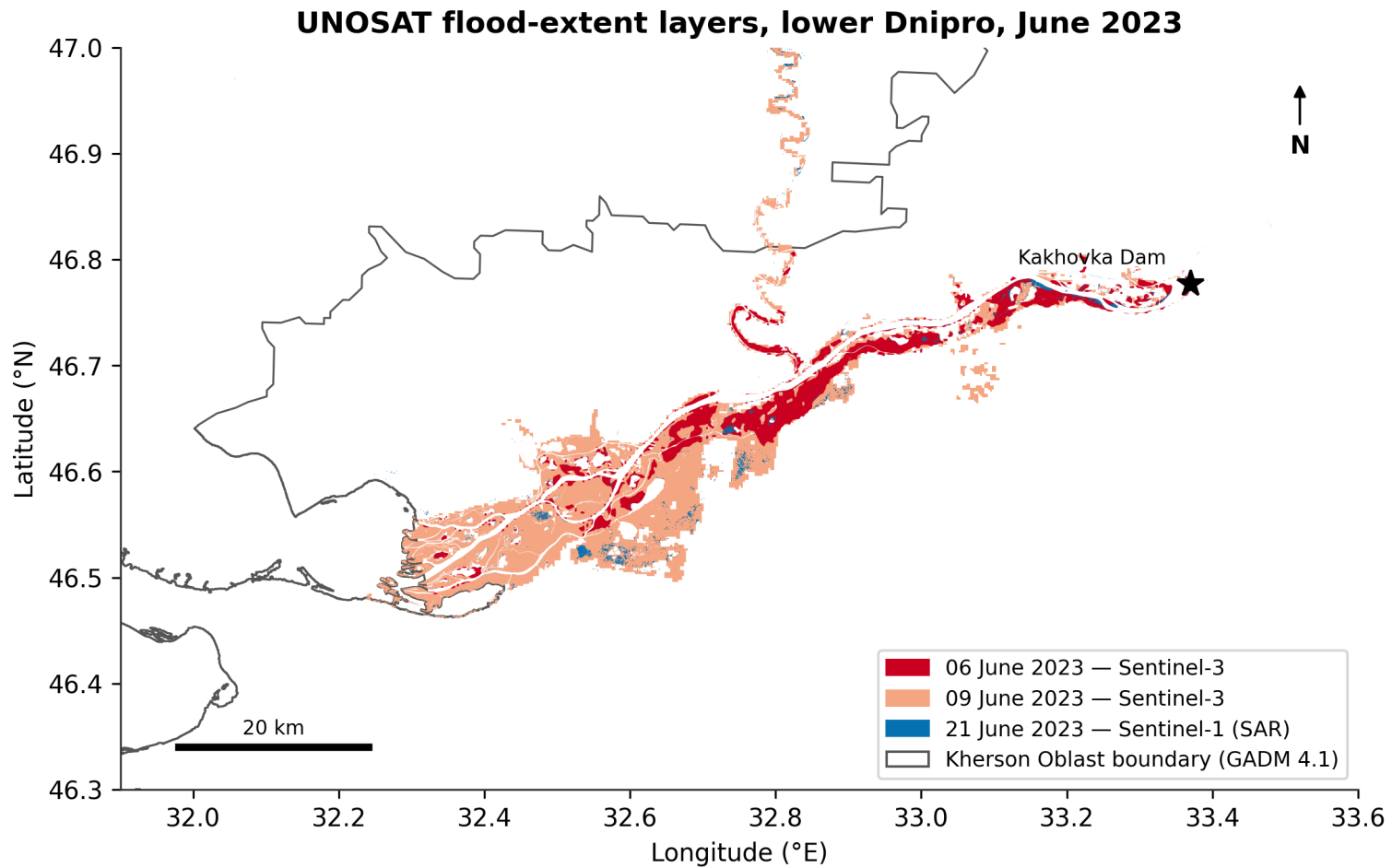
30

60 km



Same bbox and processing, July 2023. The drained reservoir tip near Nova Kakhovka is visible at top right. Context only; not used as statistical evidence.

Figure S1-4 — UNOSAT flood-extent layers

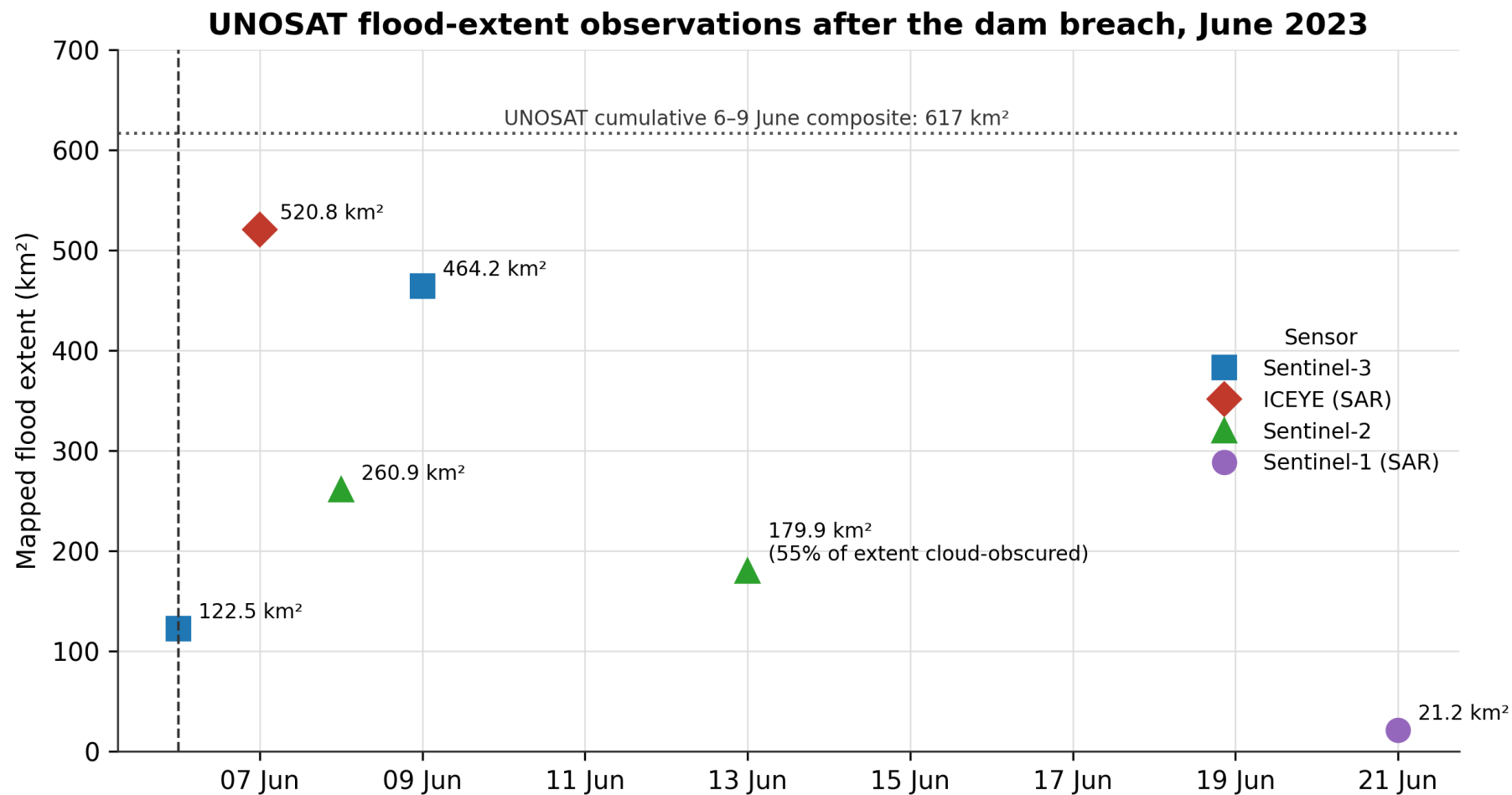


Source: UNOSAT FL20230606UKR (preliminary, not field-validated). Layers come from different sensors and analysis extents.

UNOSAT FL20230606UKR layers for 6 June (Sentinel-3), 9 June (Sentinel-3) and 21 June (Sentinel-1).

Preliminary, not field-validated; layers differ in sensor and analysis extent.

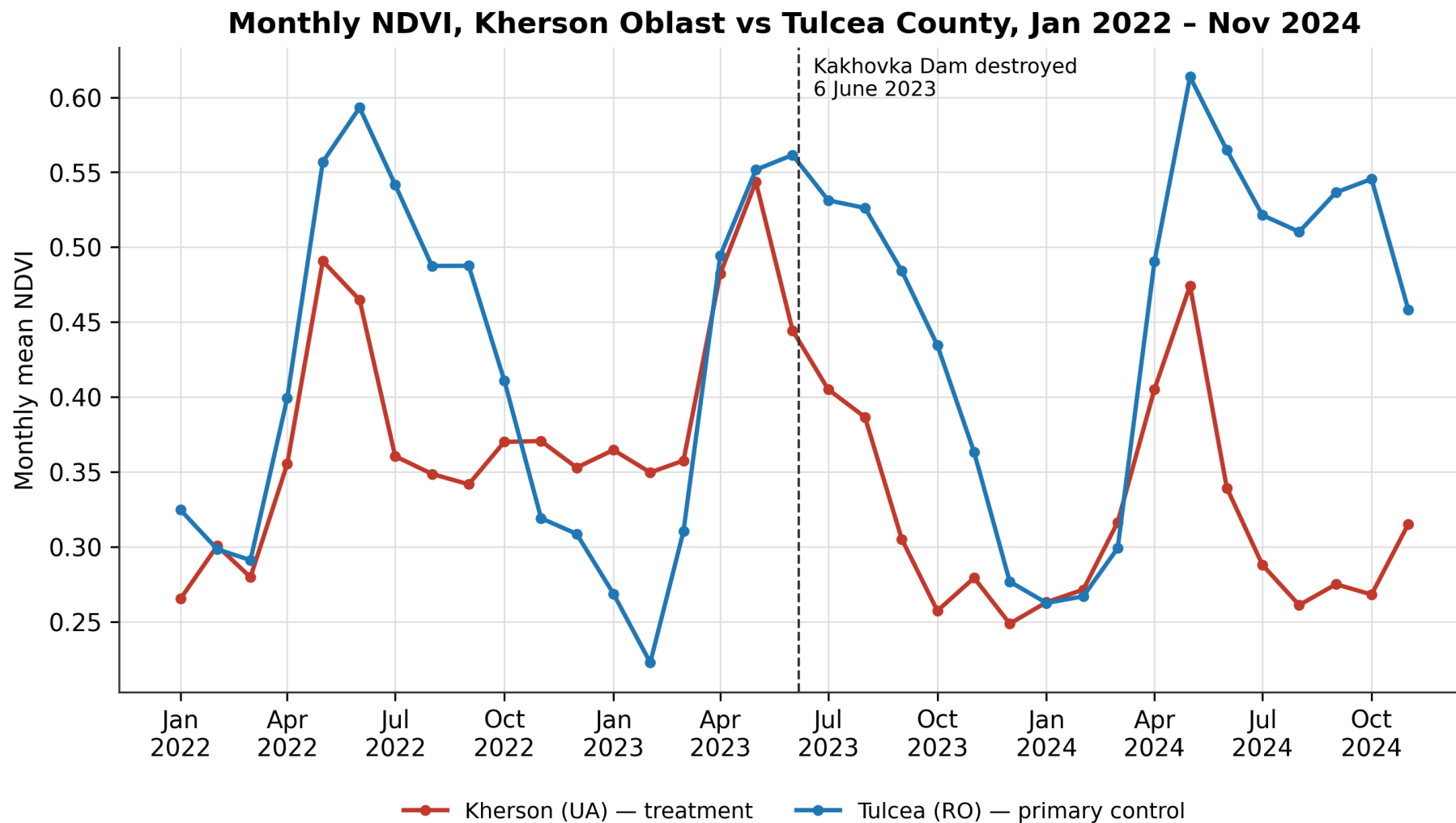
Figure S1-5 — UNOSAT flood-extent observations by sensor



Each point is a separate UNOSAT layer with its own sensor, resolution and analysis extent; values are not directly comparable across sensors.
Source: UNOSAT FL20230606UKR (preliminary, not field-validated).

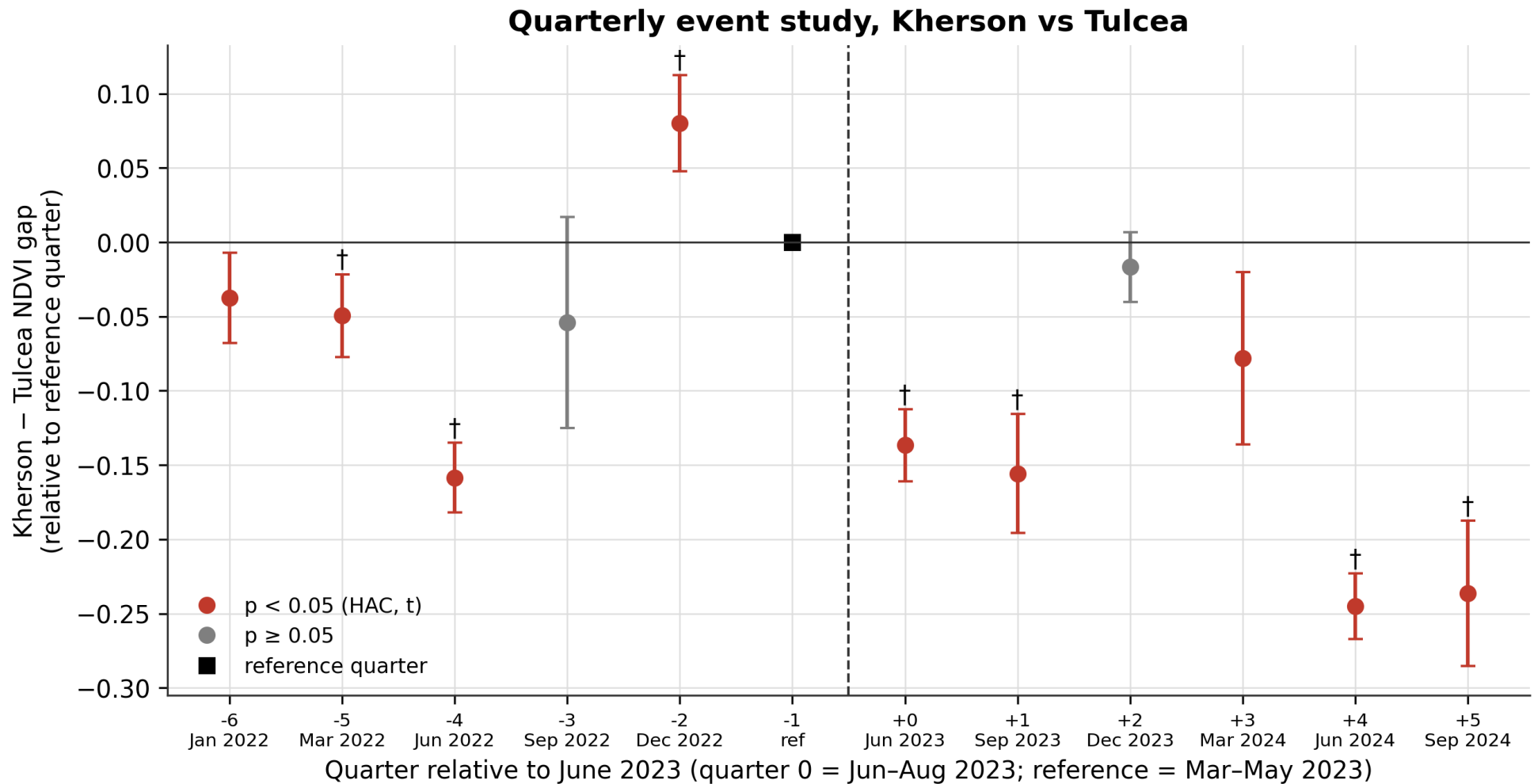
Mapped flood area in each UNOSAT layer, labelled by sensor. Separate observations, not a continuous series; see [outputs/flood_extent_table.csv](#).

Figure S1-6 — Monthly NDVI, Kherson vs Tulcea



Sentinel-2 L2A via Sentinel Hub Statistical API; GADM polygons; cloud, shadow, cirrus, snow and water masked; per-pixel monthly median on a 0.002° grid. Descriptive only.
Monthly mean NDVI (Sentinel-2, GADM polygons, SCL masking), January 2022 - November 2024. Descriptive.

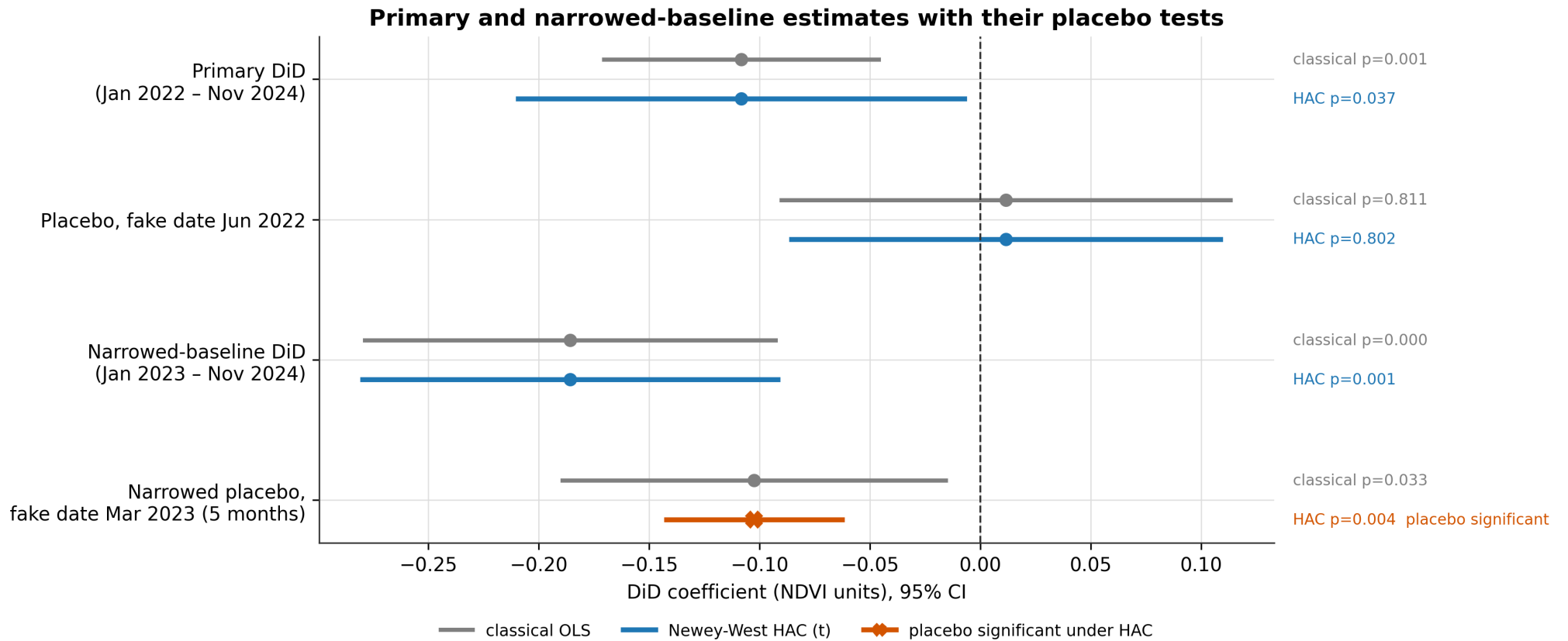
Figure S1-7 — Quarterly event study



† survives Bonferroni correction across 11 quarters ($p < 0.00455$). 95% CIs, Newey-West HAC (maxlags=3).

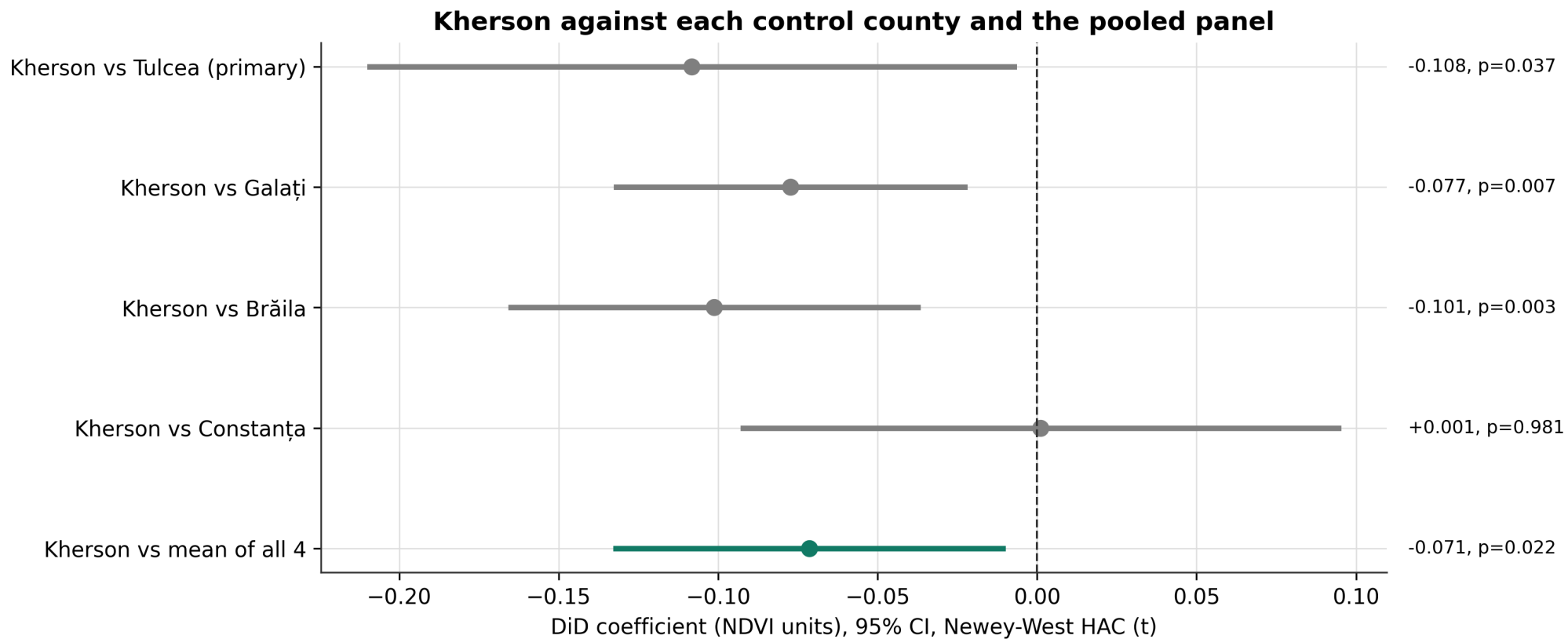
Kherson minus Tulcea NDVI gap by quarter relative to June 2023 (reference Mar-May 2023), Newey-West HAC
95% CIs; dagger marks quarters surviving Bonferroni correction.

Figure S1-8 — Primary and narrowed-baseline estimates with placebos



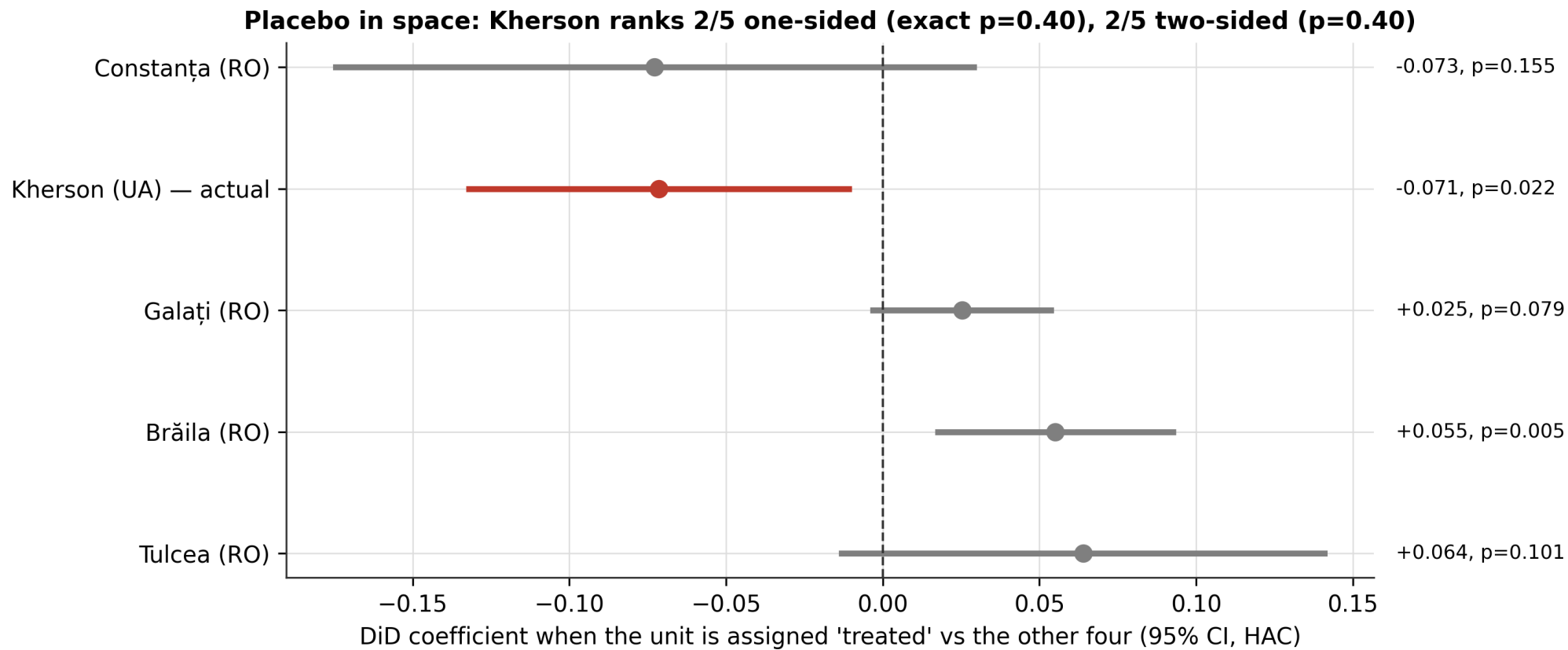
Classical and Newey-West HAC 95% CIs. Values from outputs/model_results.json.

Figure S1-9 — Kherson against each control and pooled



DiD estimates against each Romanian control county and their mean, Newey-West HAC 95% CIs.

Figure S1-10 — Placebo in space



Each of the five units assigned 'treated' in turn against the other four; exact randomization p-values from Kherson's rank.